



Visual Perception and Cognition

DSCI 454
Week 4, Lecture 2

What are we learning this lecture?



- Visual Perception and Cognition
 - How humans interpret shapes, colors, and patterns.
 - Best practices for clarity and accessibility.



How humans interpret shapes, colors, and patterns

Gestalt Principles of Visual Perception

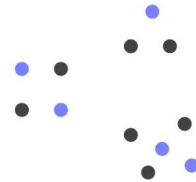


GESTALT PRINCIPLES OF VISUAL PERCEPTION

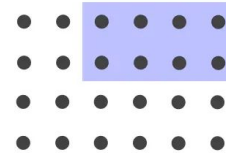
PROXIMITY



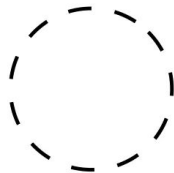
SIMILARITY



ENCLOSURE



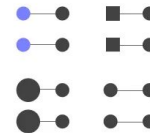
CLOSURE



CONTINUITY

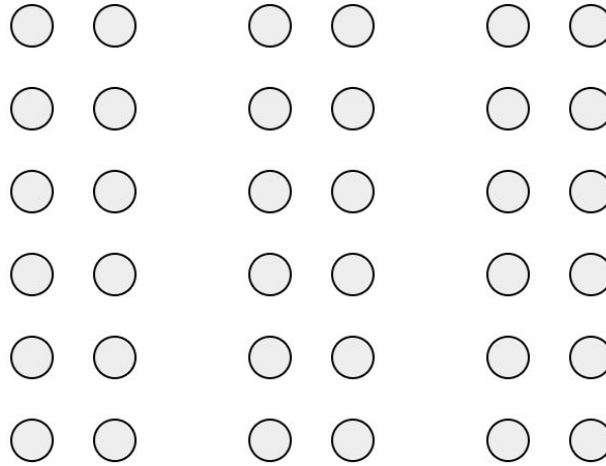
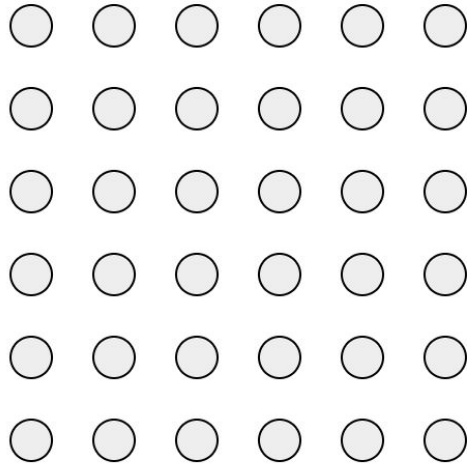


CONNECTION



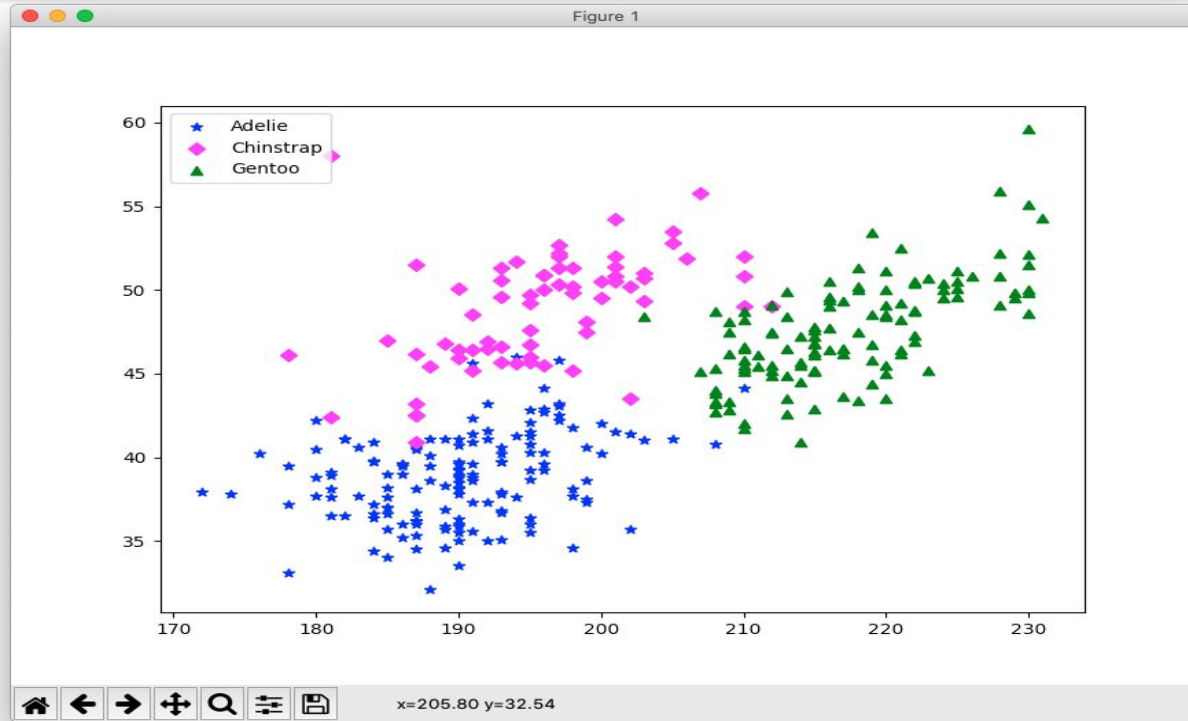
<https://nastengraph.medium.com/gestalt-principles-in-data-visualization-a4e56e6074b5>

Proximity



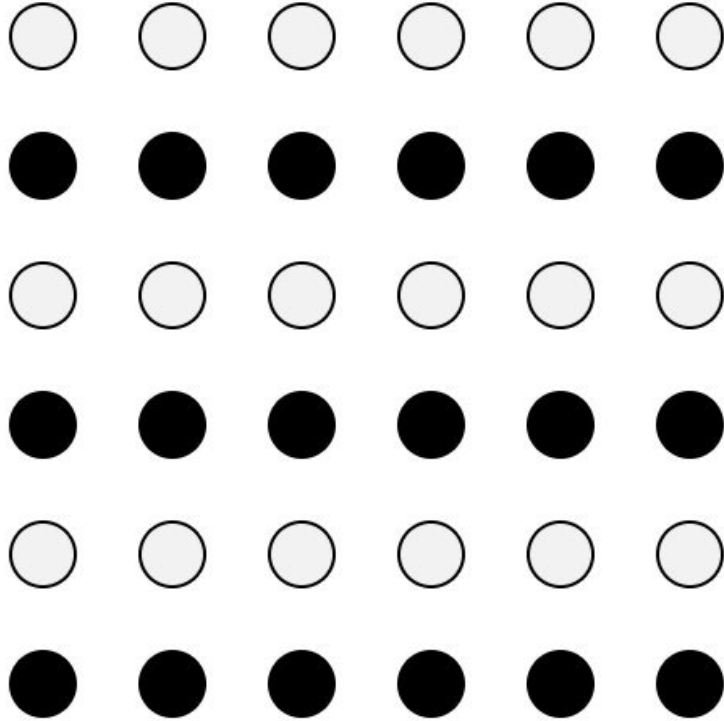
- The law of proximity: elements close to one another are perceived as a group

Proximity Example: Cluster Analysis: Penguins



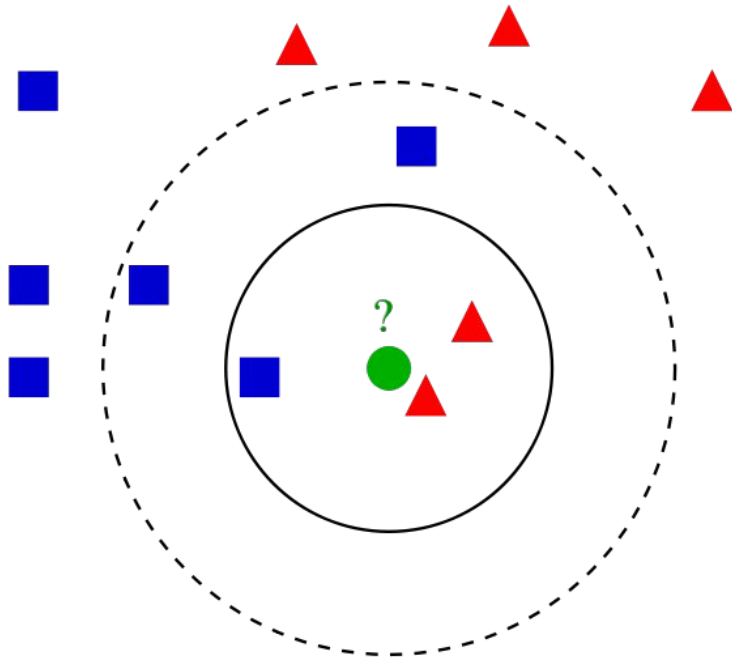
Clusters
represent
subspecies

Similarity



- Law of Similarity: Objects sharing visual characteristics are grouped together, causing:
 - The black dots appear as a single row
 - The white dots appear as a single row
- Visual Characteristics
 - color
 - shape
 - size
 - orientation

Similarity Example: KNN

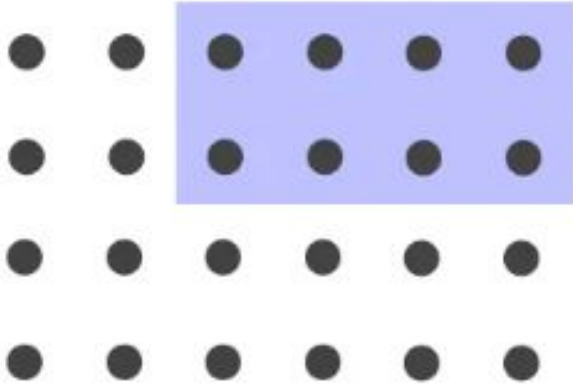


source: https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

- In AI/ML, K-Nearest Neighbors (KNN) groups similar characteristics near each other.
- Note: It's a non-visual algorithm but is often visualized like this during teaching/learning



ENCLOSURE

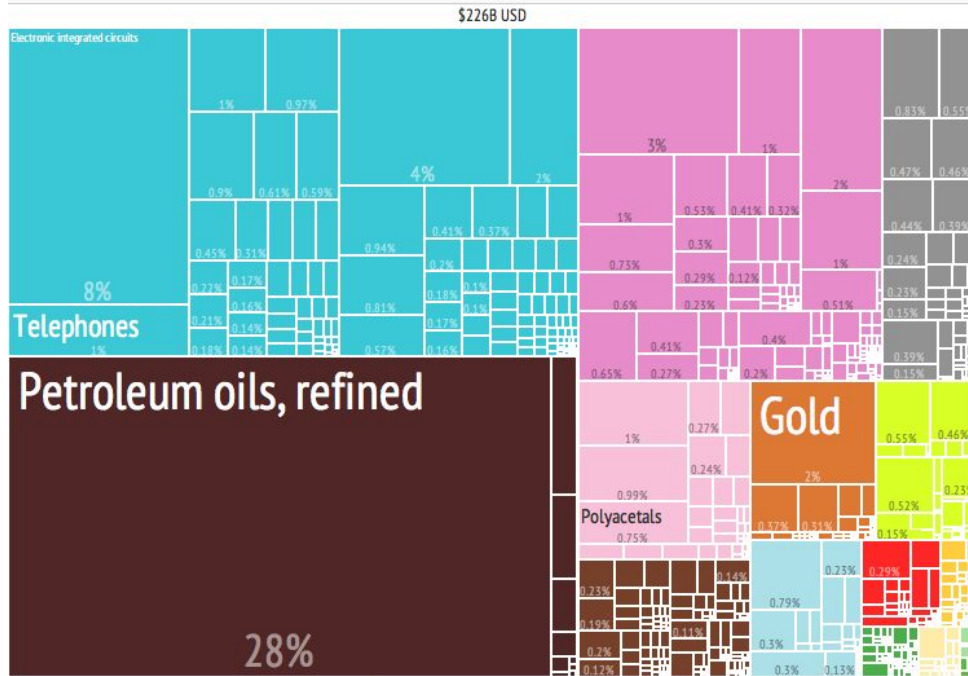


- Law of Enclosure: Objects physically enclosed together are grouped together



Enclosure Example: Treemaps

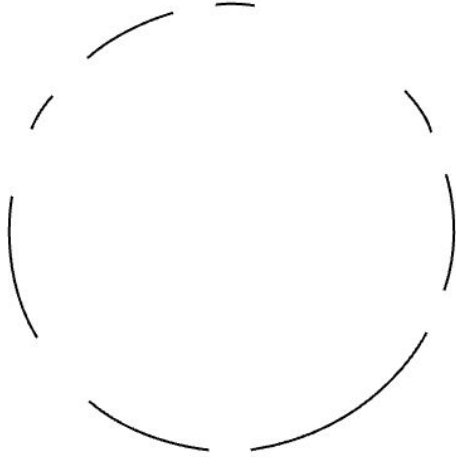
What did Singapore export in 2012?



- Colors and borders both group

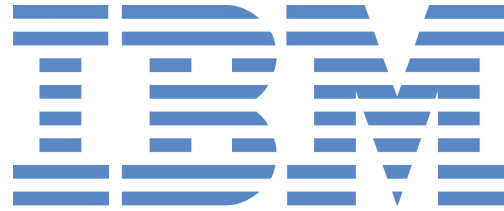
By Gordon.silvermanaz - Own work, CC BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=34068665>

Closure



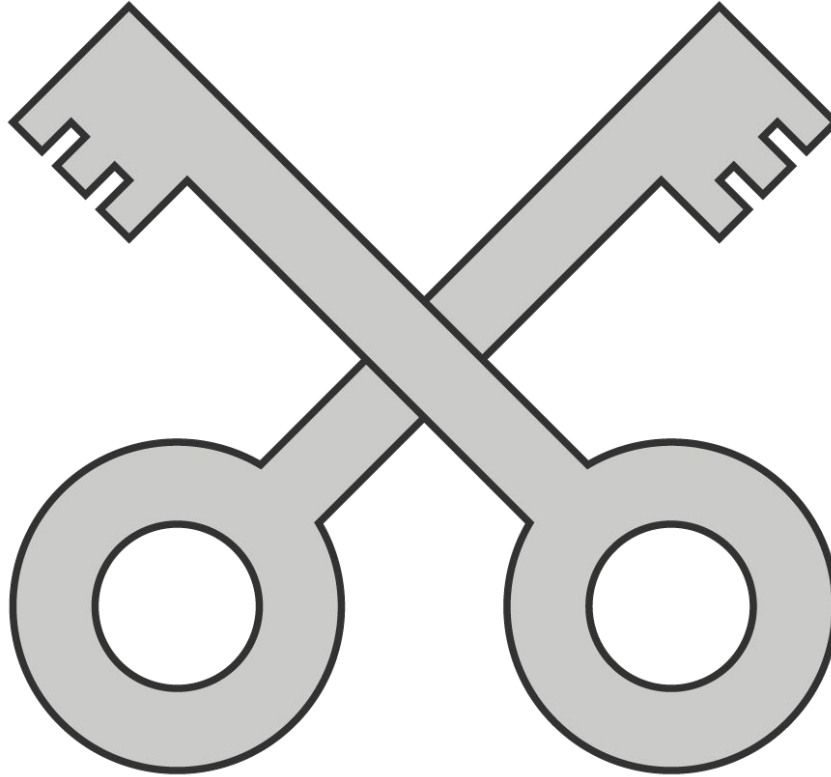
- Law of Closure: Incomplete figures are perceived as whole shapes
- The human brain is very good at interpolation

Closure Examples



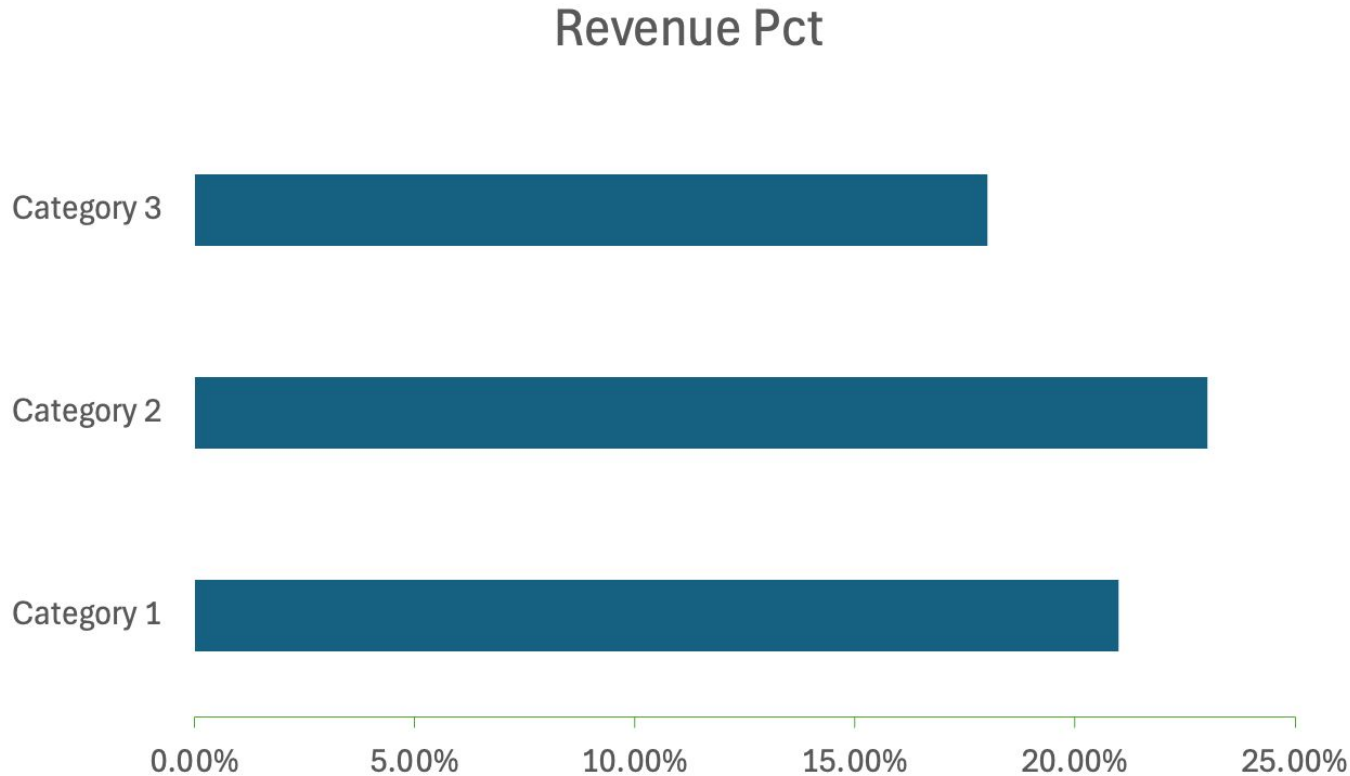
- IBM logo
- PBS logo
 - 3 heads
- FedEx
 - Arrow
- Use / balance of negative space w/ positive space

Continuity



- Law of Continuity: Intersecting objects are perceived as uninterrupted entities
- Background key appears continuous
- e.g. Car drives under an overpass

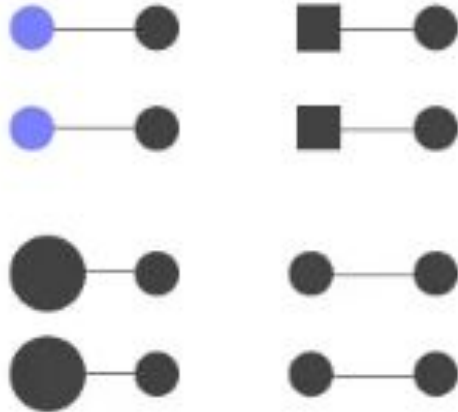
Continuity Example



- Note the y-axis line is completely removed yet our eyes are able to follow just fine



CONNECTION

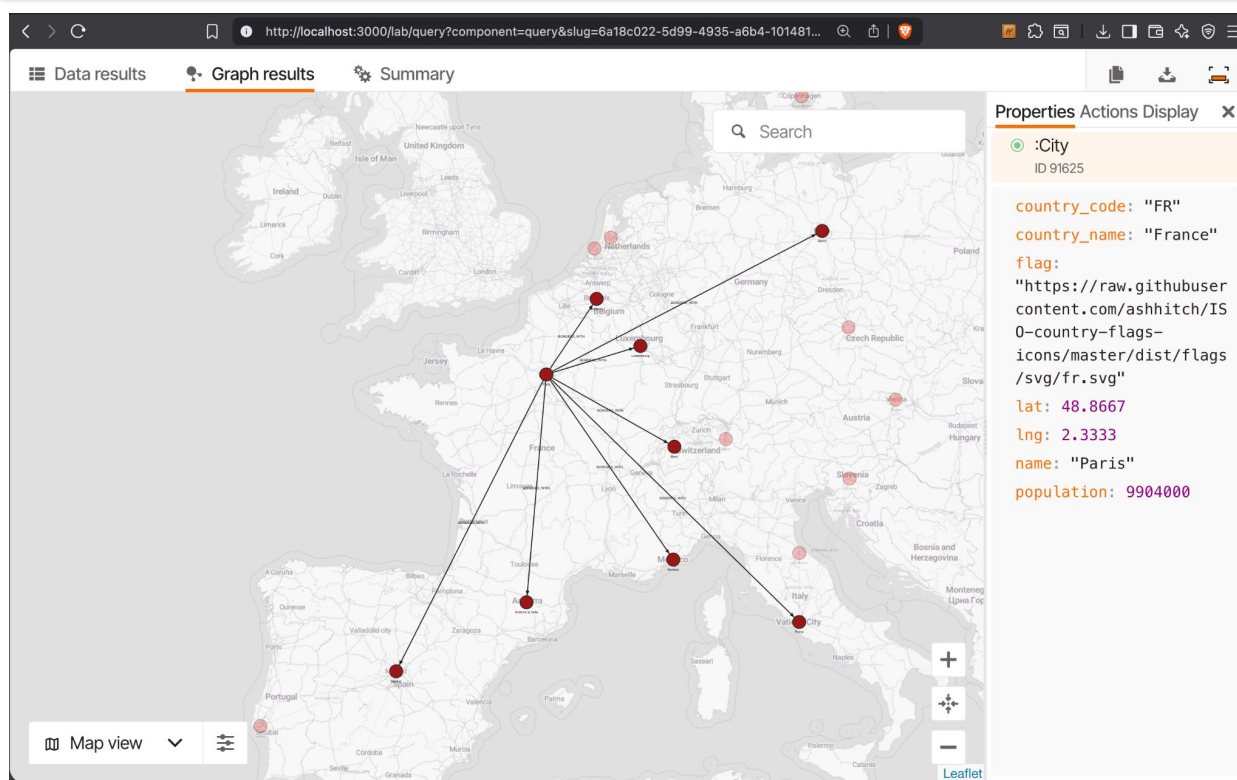


- Law of Connection: Objects physically connected are grouped together **overriding** other visual characteristics
 - color
 - shape
 - size
 - orientation

Connection Example



- Knowledge Graphs
 - Neo4j
 - memgraph



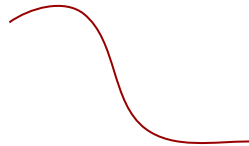


Best practices for clarity and accessibility



Clarity Factors to consider

- Every chart should have:
 - Title, Axis labels, Units of measure (\$, mi. m/s, mph, etc.)
 - data labels, legend, as applicable
- Language
 - Do not use overly complicated academic words
- Alignment
- Color in general
 - Busy
- *Flow*
- Continuity
 - Eye movement
 - Cognitive load
- Show comparisons **side by side**
 - Same logic as bar over pie charts
- Present **mul-ti-ple options** for discussion / feedback
 - It's the presentees decision
- **Whitespace:** Be cognizant of / utilize / balance / deliberate
 - i.e. Powerful silent pause during a speech, e.g. Steve Jobs
 - Anti-Clutter



Accessibility Factors to consider



- Color blindness
 - Red vs green hues/shades
 - 8% of males, half percent of females
 - Color palettes (seaborne built-in “colorblind” palette)
- Alt Text
 - Visual description used by screen readers
- Hover over text
 - Do not rely on this alone since not everyone is using a mouse
- *Font*
 - *Typeface and Size*
 - `Serif` vs `Sans Serif`

Questions?

